Regional Economic Disparities in Denmark

When Did Hovedstaden Pull Ahead?

by Valeria Angelino Verano

1. Introduction

Throughout the EOSE09 lectures, we have heard about regional disparities in countries like Sweden, Italy, and the United States. However, I always wondered why Denmark was never mentioned. This led me to carry out a bit of research to find out if Denmark would be a good pick for my lab paper, and I was pleased to find that writing this paper on Denmark would be an interesting but achievable task.



Adapted from: Region Nordjylland (2025)

Figure 1. Map of Denmark and its regions

Denmark is a small country located in central Europe composed of 5 regions: Nordjylland, Midtjylland, Southern Denmark, Sjaelland, and Hovedstaden, as shown in *Figure 1*. According to Christiansen et al. (2020), Denmark runs one of the world's biggest regional government sectors in which all regions work to achieve national growth. Each region individually manages the majority of Denmark's welfare system including daycare, primary schools, healthcare, etc. (Christiansen et al., 2020). Thus, each region is responsible for its regional development with regional financing of 100 billion Danish kroner to split amongst the provision of health and social services, business promotion, tourism, employment, public transport, etc (Hansen & Holst, 2012). According to the 2022 OECD report *Regions and Cities at a glance: Denmark*, there are regional disparities amongst Danish regions in terms of employment, productivity, population density, and economic growth, with regions like Hovedstaden overperforming regions like Midtjylland and Southern Denmark.

So, when and why did Hovedstaden pull ahead of other regions? Do the trends point to an eventual convergence? This lab paper aims to explore the answers to these questions by looking at the economic regional path of the five regions of Denmark, specifically looking into variables such as the regional GDP per capita, employment share, and regional growth between 1900 and 2015. The data was obtained via the Rosés & Wolf database (2020), and it was imported into the software STATA for data analysis and visualization.

2. Methods & Limitations

For this paper, I used the 2020 version of Rosés-Wolf database which was imported into the software STATA. The code used for this paper has been included as a separate txt file titled "Lab Paper Code". To understand the variables used, the command *tabstat* was used extensively to find measures of central tendency including min, max and mean. This allowed me to choose fitting brackets and bin widths for maps, and fitting scales for graphs. Finally, all figures created are found and discussed in the *Results & Analysis* section.

It should be noted that the fact that Denmark only has five regions becomes a challenge in the calculation and analyses of both beta-convergence and sigma-convergence. This is due to the fact

that a small sample size can increase the variance in estimates and dispersion measures, potentially making it more difficult to identify convergence patterns.

3. Results & Analysis

a. Regional Disparities and Trends

Figure 2 shows the regional GDP per capita of the five Danish regions from 1900 to 2015. From this figure, it can be seen that all regions had a similar GDP per capita (GDP pc) in 1900, with little differences over the next 40 years. However, we see a big increase in Hovedstaden GDP per capita from 1940 to 1950. This difference seems to be equalized in 1960, with all regions having similar GDP per capita. Despite this, from 1960 onwards, we can see Hovedstaden's GDP per capita going higher than any other region, with the gap widening from 1960 to 2015. Overall, *Figure 2* suggests that 1950 and 1960 marked the beginning of the regional disparities in Denmark.

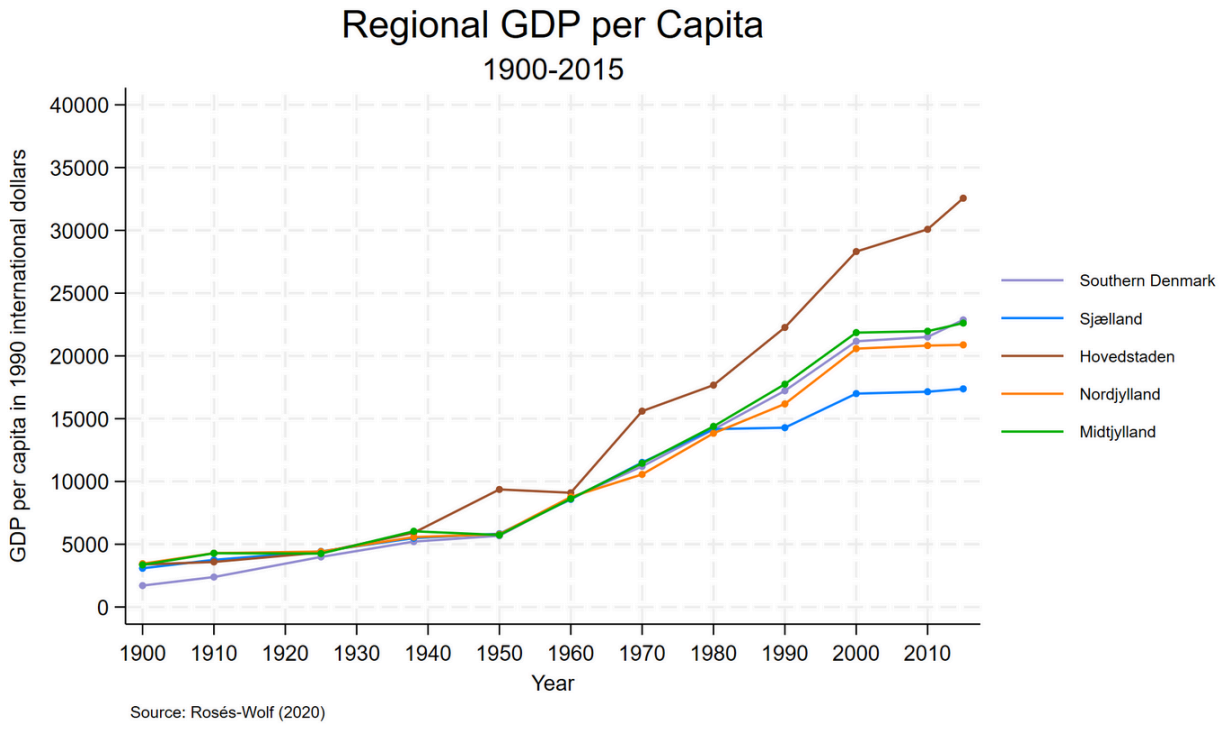


Figure 2. Regional GDP per capita

One of the explanations for the divergence between Hovedstaden and other Danish regions could be the employment share and population density. According to Hansen & Holst (2012), by 2010 more than 30% of the total Danish population was found in Hovedstaden, which is the smallest region by area (Hansen & Holst, 2012). *Figure 3* shows the population density of the five Danish regions from 1900 to 2015. From *Figure 3*, we can see that Hovedstaden has had the highest population density out of the five regions since 1900. Hovedstaden's population density doubled from 1900 to 1960 and continued to increase up until 2015 suggesting an increasing urbanization. The population density of the remaining four regions averaged around 100 inhabitants per kilometer square, which could suggest little population growth or even little urbanization. This becomes significant as high population density could lead to agglomeration effects in which firms are workers benefit from close proximity increasing innovation, knowledge spillovers (Rappaport & Sachs, 2003). This accumulation of human capital could also suggest an accumulation of infrastructure. If this is the case, Hovedstaden would become more attractive for businesses, workers, and continued investment and growth, as pointed by Davis & Weinstein (2002). These two factors could increase Hovedstaden's persistence, in which Hovedstaden would keep its advantages over time and achieving long-term regional growth.

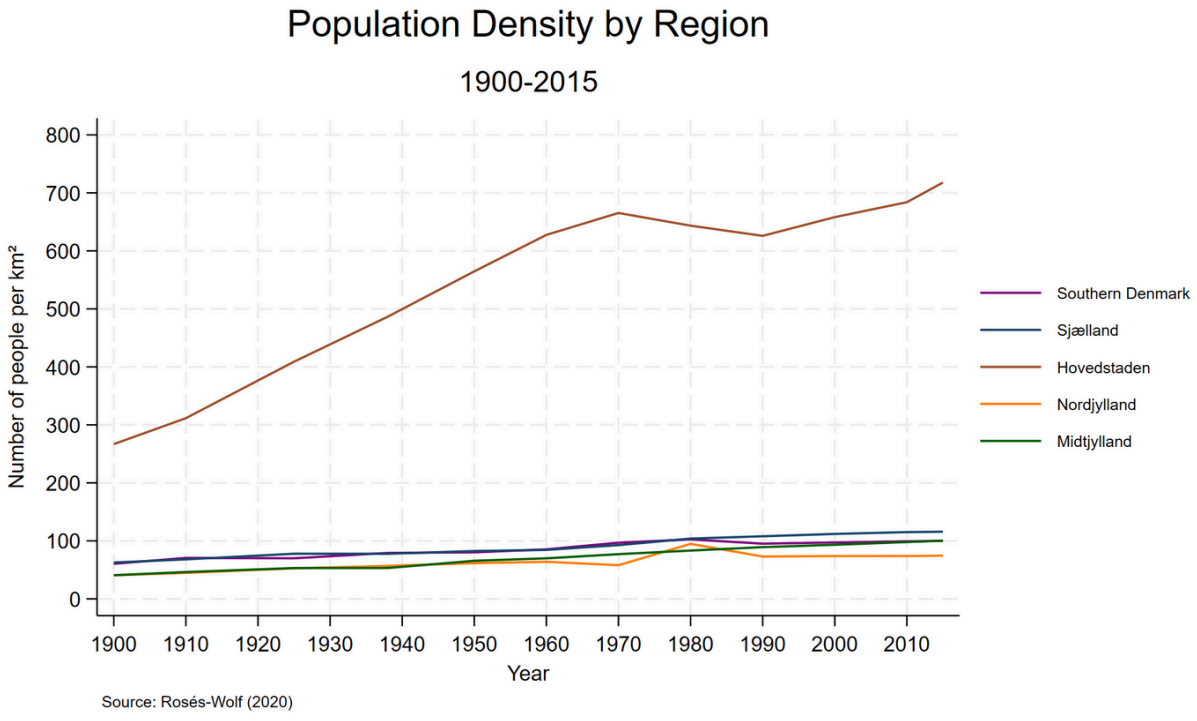


Figure 3. Population Density by Region

For a better insight into urbanization and the changes in regions, *Figure 4* visually shows the share of employment amongst agriculture, industry, and services in Denmark by region comparing 1950, 1960, and 1970. The first row of maps shows the share of employment in agriculture where we can see all regions had a relatively high focus on agriculture, except for Hovedstaden, which had less than 10%. The share of agriculture in Denmark decreased by 1960, and ultimately most regions had between 10 and 20% share in agriculture - Hovedstaden remained under 10%.

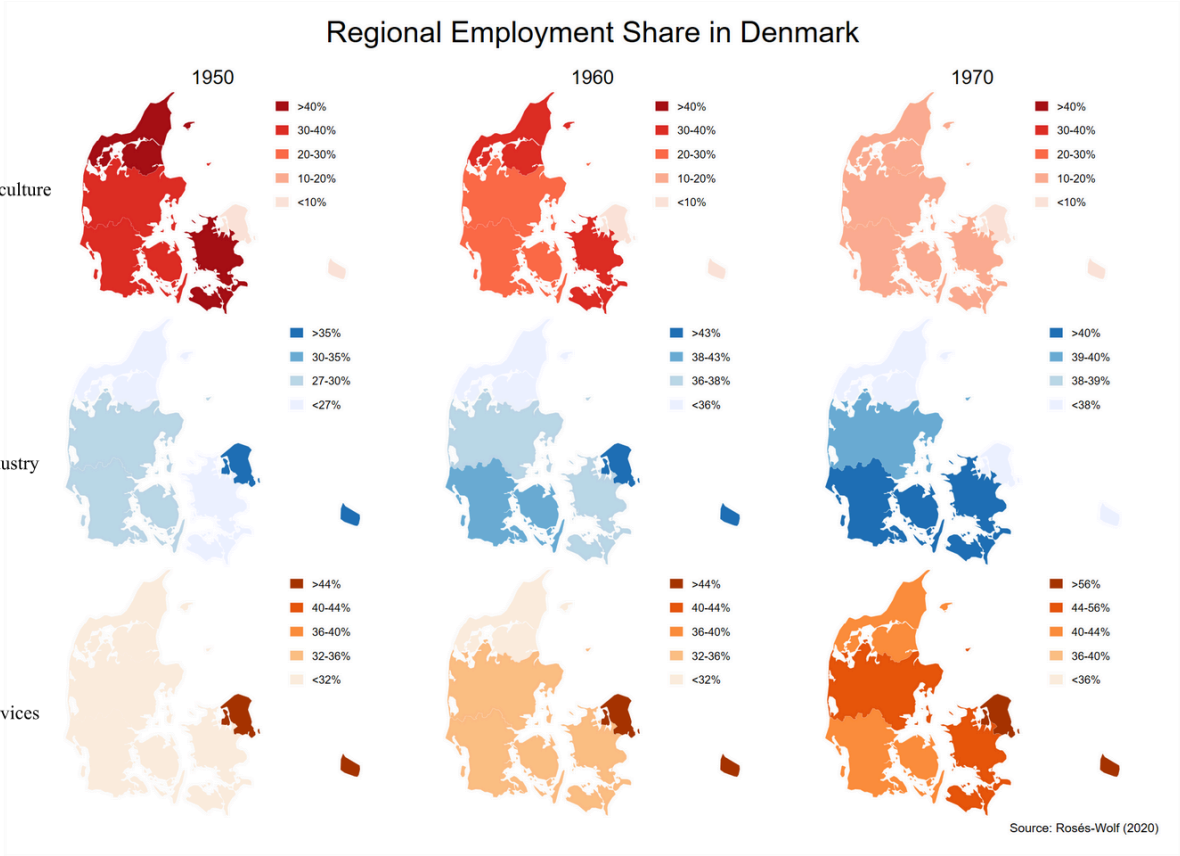


Figure 4. Regional Employment Share amongst Danish Regions

In contrast. We can see the second row of maps that show the share of employment in industry and manufacturing. This gives a good indication of Hovedstaden's early and higher industrialization and urbanization compared to the other regions as in 1950, Hovedstaden had a share higher than 35%, with the other regions having less than 30%. By 1960 other regions increased their employment share in industry to the point that by 1970 Hovedstaden had the lowest share of employment in industry. Lastly, the last row shows the share of employment in

services. In 1950 Hovedstaden had a share of higher than 44% in services, whilst other regions had less than a third. By 1960 the services share began increasing in the other regions, with Hovedstaden still having the highest share. By 1970 More than 56% of Hovedstadens employment was in the services sector whilst other regions remained between 40 and 56%.

From *Figure 4* we can see that Hovedstaden switched from agriculture to industry before other regions, and even moved into the services sector by 1950. These findings align with those of *Figure 3*, as Denmark boosted industrialization in the 1950s and therefore higher population density amongst more urbanized regions. Further, DenmarkHistorien (2025) points out that after World War II and as part of the European Recovery Program, Denmark received a 300 million USD money injection to improve the efficiency of agriculture and develop the manufacturing industry. Moreover, the International Bank for Reconstruction and Development (1963) shared that the Danish economic expansion began around 1958 as the result of government policies promoting industrialization. The combination of money injection and government policies could explain the trends in industry and agriculture share from *Figure 4*.

The move from agriculture to industry was accompanied by internal migration from the countryside to towns and more urbanized regions (such as Hovedstaden) since by 1970, only one-fifth of the Danish population lived in rural areas (International Bank for Reconstruction and Development, 1963). Although it is not possible to see the extent of internal migration in *Figure 3,* we can still assume that the increase in population density in Hovedstaden between 1950 and 1970 was the result of the economic expansion described above.

a. On the Road to Convergence?

Regional disparities could increase or decrease over time and to find out Denmark's trends between 1950 and 2015, I created a Beta-Convergence with the *log of initial GDP per capita* in the X-axis, and *average growth rate* in the Y-axis, shown as *Figure 5* below. According to Barro & Sala-i-Martin (1991) regions with a low initial GDP per capita tend to grow at faster rates than wealthier ones, which would be reflected in a negative slope when plotting. However, from *Figure 5* we can see that this is not the case for the Danish regions between 1950 and 2015. Firstly, we see that we have a positive slope in the Beta-Convergence which suggest divergence. We can see that regions with low initial GDP per capita - such as Sjaelland - do not have a higher

growth rate. Although regions with low initial GDP per capita like Southern Denmark and Midtjylland have experienced high growth rates, the region of Hovedstaden has also a high growth rate with a high initial GDP per capita. From *Figure 5* we have no indication that poorer regions are catching up, meaning no convergence.

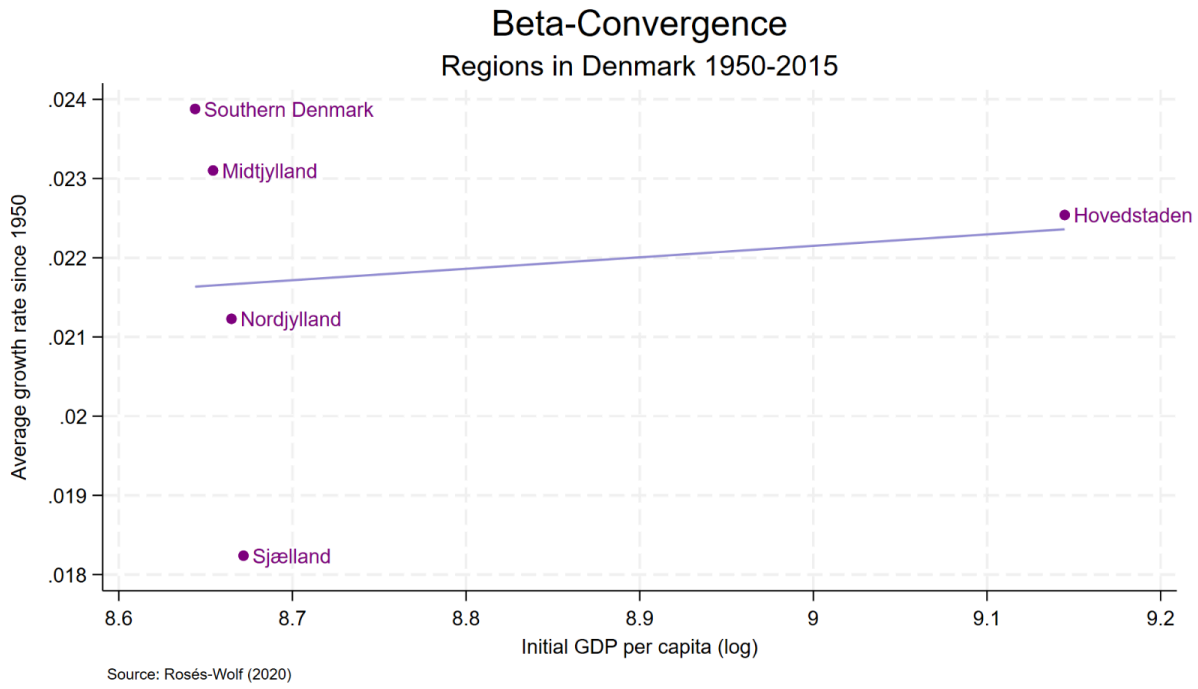


Figure 5. Beta-Convergence

I also decided to look into sigma-convergence to identify trends in the dispersion of GDP per capita amongst danish regions, shown in *Figure 6*. *Figure 6* shows both the trend for weighted and unweighted coefficient of variation. This helps to understand wether population distribution influences regional economic disparities which could be significant given the findings in *Figure 3*. However, the difference in the weighted and unweighted coefficient of variation in the case of Denmark is very little. From *Figure 6* we can see that the coefficient of variation starts high around 1900, which decreases significantly and meets again in 1950, indicating that economic disparities amongst regions fluctuated between that period. This aligns with the findings in *Figure 2*, where Hovedstaden pulled ahead momentarily. Around 1980 we can see that the coefficient of variation increased continuously pointing to increasing regional economic disparities. This could be the long-term effects of the government policies and industry boom from the 1950s which led to the urbanization and development of Hovedstaden, and the internal

migration from other regions. Ultimately, *Figure 6* does not point to convergence amongst the Danish regions.

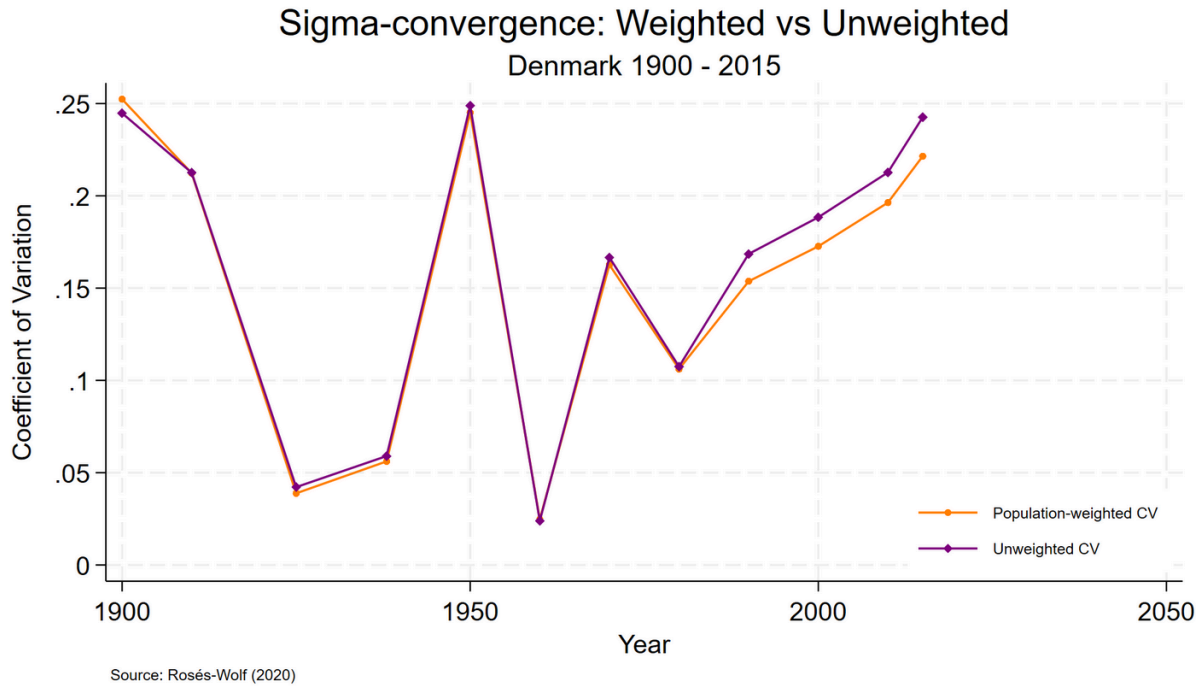


Figure 6. Sigma-Convergence

These divergence trends bring up the term of persistence pointed when discussing *Figure 3*. Hovedstaden may have regional advantages that will persist over time, making them likely to remain ahead of the other Danish regions. As pointed earlier, Hovedstraden seems to experience infrastructure accumulation and agglomeration (and in turn, knowledge spillovers), which are two factors that could explain its persistence and continuous economic growth. As pointed by Daalgard et al (2022) there are factors that could improve the development of disadvantaged regions including infrastructure development, human capital development, and research & innovation. Ultimately, it is clear that Hovedstaden will continue to grow and regional disparities will persist.

4. Conclusion

This paper allowed me to find answers to the questions posed in the introduction *When and why did Hovedstaden pull ahead of other regions?* And *Do the trends point to an eventual convergence?* As shown in the *Results & Analysis* section, it is clear that Hovedstaden pulled ahead between 1940 and 1960. This period marked the beginning of the divergence of Hovedstaden over the remaining regions which can be attributed to an early shift to industry. We saw that Denmark aimed for economic growth as posed by the government policies and the foreign aid received during the period. Hovedstaden seems to be the region that benefited the most from a shift to industry and a soon shift to services. We found that Hovedstaden is the most densely populated region, but we also found that it has been the case since the 1900, reaching its peak after the internal migration from the 1950s. It makes sense that the most densely populated region urbanizes first and faster than those less populated given agglomeration effects. Hovedstaden seems to be a region whose regional advantages have persisted over time, and will continue to achieve long-term economic growth.

The economic regional disparities in Denmark do not appear to be closing up. The data suggests that Denmark experienced a period of economic convergence until around 1950, and this was followed by an increasing divergence from 1950 onwards. This has led to Hovedstaden continuous economic advantage, as it remains the region with the highest GDP per capita, and the data points that regional economic disparities will continue in Denmark.

Despite the findings of this paper, the regional economic disparities of Denmark compared to a country like Italy, are minimal. This could be due to the regional government sector that Denmark has. Given that each region manages its own budget, it can become easier for a region to identify its points of growth and invest in those sectors that could be improved. Although not covered in this paper, a further research into how each region spends its budget would be interesting to understand further regional disparities, their roots, and possible solutions.

5. References

Arbejdermuseet (2025). *The 1950s*, Accessed through: https://www.arbejdermuseet.dk/en/whats-on/exhibitions/the-1950s/ [Accessed 14th March 2025]

Barro, R., Sala-i-Martin, X. (1991). Convergence across States and Regions, *Brookings Papers on Economic Activity*, vol. 22, no. 1, pp 107-182, Available online: https://doi.org/10.2307/2534639

Dalgaard, C., Kaarsen, N., Olsson, O., & Selaya, P. (2022). Roman roads to prosperity: Persistence and non-persistence of public infrastructure, *Journal of Comparative Economics*, vol. 50, no. 4, pp. 896-916, Available online: https://doi.org/10.1016/j.jce.2022.05.003

Daugbjerg, M. (2014) Borders of belonging. experiencing history, war, and nation at a Danish Heritage Site. Berghahn Books (Museums and collections: volume 5). Available online: https://search-ebscohost-com.ludwig.lub.lu.se/login.aspx?direct=true&AuthType=ip,uid&db=cat02271a&AN=atoz.ebs2705281e&site=eds-live&scope=site [Accessed 30th January 2025]

Davis, R, Weinstein, R. (2002) Bones, Bombs, And Break Points: The Geography Of Economic Activity, *American Economic Review*, vol. 92, pp. 1269-1289, Available online: https://www.aeaweb.org/articles?id=10.1257/000282802762024502

Hansen & Holst, 2012. The Regions in Brief, [pdf] Available online: https://www.regioner.dk/media/1334/regionerne-kort-fortalt-2011-engelsk.pdf [Accessed 10th February 2025]

International Bank for Reconstruction and Development. (1963) The Economy of Denmark [pdf] Available online: https://documents1.worldbank.org/curated/en/382811468244508636/pdf/multi-page.pdf [Accessed 12th March 2025]

OECD. (2022). Regions and Cities at a glance: Denmark [pdf], Available online: https://www.oecd.org/content/dam/oecd/en/publications/reports/2024/11/oecd-regions-and-cities-at-a-glance-2022-country-notes_00a3f24b/denmark_09415a55/77d45c4d-en.pdf [Accessed 12th March 2025]

Rappaport, J., Sachs, J. (2003). The United States as a Coastal Nation, *Journal of Economic Growth, vol. 8, pp.* 5–46, Available online: https://doi.org/10.1023/A:1022870216673

Region Nordjytland (2025). *Regions in Denmark*, Accessed through: https://rn.dk/Service/English [Accessed 29th January 2025]

Humboldt-Universität zu Berlin (2020). Rosés-Wolf database on regional GDP (version 6, 2020), Accessed through: https://www.wiwi.hu-berlin.de/de/professuren/vwl/wg/roses-wolf-database-on-regional-gdp [Accessed 29th January 2025]